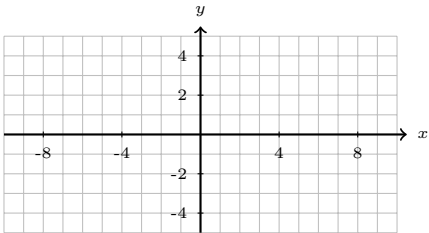


Part 1: Parent Functions - Complete the tables and sketch the graphs.

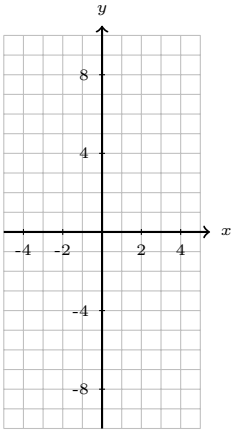
2. $f(x) = x^3$

1. $f(x) = \sqrt[3]{x}$

x	y
-8	
-1	
0	
1	
8	



x	y
-2	
-1	
0	
1	
2	

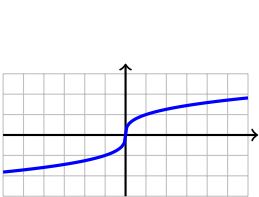


Intercept Reminder: To find the x -intercept, set $y = 0$ and solve. To find the y -intercept, set $x = 0$ and solve.

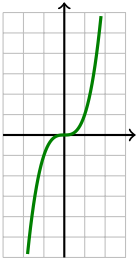
Part 2: Identify the key features of each parent function.

	3. $f(x) = \sqrt[3]{x}$ (Cube Root)	4. $f(x) = x^3$ (Cubic)
Domain		
Range		
Intercept(s)		
End Behavior	As $x \rightarrow \infty, y \rightarrow$ _____	As $x \rightarrow \infty, y \rightarrow$ _____

Part 3: Which function is it? Write C (Cubic) or CR (Cube Root).



5. _____



6. _____



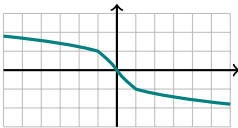
7. _____



8. _____



9. _____



10. _____



11. _____



12. _____

Part 4: The Big Distinction — Fill in the blanks.

	Cube Root $\sqrt[3]{x}$	Cubic x^3
13. When x gets BIG...	y grows slowly (Wide)	y grows fast (Tall)
14. Shape	_____ “S”	_____ “S”

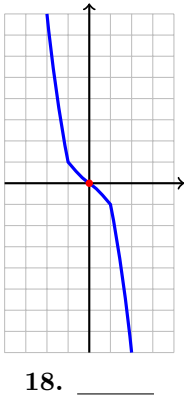
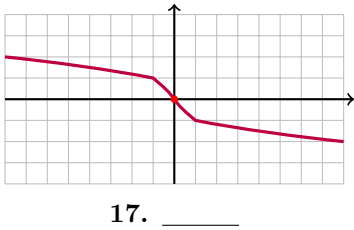
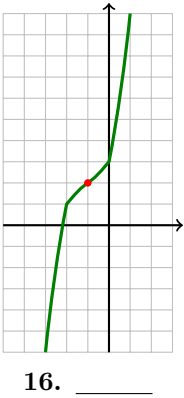
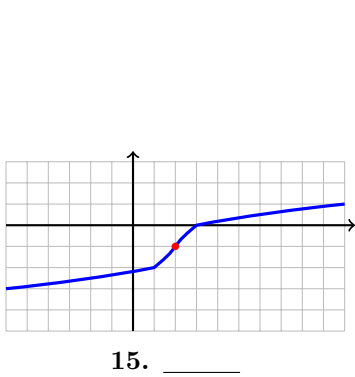
Part 5: Match the Equation to the Graph. Find the intercepts.
Look at the shape (Tall vs Wide) AND the "Center Point" to match them!

A) $y = (x + 1)^3 + 2$

B) $y = \sqrt[3]{x - 2} - 1$

C) $y = -x^3$

D) $y = -\sqrt[3]{x}$



Find the x - and y -intercepts for each equation above:

	A) $y = (x + 1)^3 + 2$	B) $y = \sqrt[3]{x - 2} - 1$	C) $y = -x^3$	D) $y = -\sqrt[3]{x}$
x -int				
y -int				